

Electronic Sales Data: Customer Purchase Behaviour

| Trinity College Dublin
| Advanced Topics in Analytics
| MSc Business Analytics



Contents

- 1. Introduction.....2**
- 2. Methodology..... 2**
 - 2.1. Data Preparation and Feature Engineering..... 2
 - 2.2. Model Development.....3
- 3. Analysis and Results..... 5**
 - 3.1. RQ1: The Effect of Loyalty Membership on Order Value..... 5
 - 3.2. RQ2: Determinants of Product Ratings..... 7
 - 3.3. RQ3: Price Elasticity and Add-On Influence.....10
- 4. Discussion and Conclusions..... 13**
 - 4.1. Summary of Key Findings..... 13
 - 4.2. Project Management Prioritisation..... 14
 - 4.3. Ethical Analytics & Limitations..... 15
 - 4.4. Conclusion..... 16
- 5. Sources..... 17**
- 6. Appendix..... 17**
 - 6.1. RQ1:..... 17
 - 6.2. RQ2:..... 18
 - 6.3. RQ3:..... 20
 - 6.4. Key Findings:..... 22

1. Introduction

Understanding the drivers of customer purchasing behaviour is central to effective decision-making in retail and e-commerce. Firms routinely invest in loyalty programmes, logistics improvements, and pricing strategies with the expectation that these factors will influence spending and satisfaction. However, the extent to which these interventions are effective is not always clear, particularly when customer behaviour is shaped by multiple interacting factors. This study analyses a transactional dataset from an electronics retailer from September 2023 to September 2024, to examine three related questions focused on understanding the drivers of order values (\$ amount per order) and customer value:

- **RQ1:** How do sociodemographic factors associate with loyalty membership, and what is the causal effect of membership on the purchase amount?
- **RQ2:** Which product and transaction characteristics (e.g., product type, order status, shipping type) affect product ratings?
- **RQ3:** How do product prices and add-ons influence final prices, and does price sensitivity vary across product types?

To address these questions, a combination of econometric approaches is employed. Propensity Score Matching (PSM) is used to account for selection bias in evaluating the effect of loyalty membership; a multinomial logistic regression model is applied to analyse the determinants of customer ratings; and a Two Squared Least regression (2SLS) and log-log regression models are used to estimate price elasticity and assess heterogeneity across product types. The findings contribute to a better understanding of how pricing, customer programmes, and operational factors interact in shaping purchasing behaviour. In doing so, the analysis provides insights that are relevant for both managerial decision-making and the interpretation of customer data in applied settings.

Note: the terms “order value”, “purchase amount”, “spending”, and “expenditure” are used interchangeably throughout this assignment.

2. Methodology

2.1. Data Preparation and Feature Engineering

The data was preprocessed by identifying data types and missing values. Binary categorical columns (**Gender**, **Loyalty Member**, **Order Status**) were binary-encoded. Multi-category nominal columns (**Product Type**, **Shipping Type**, **Payment Method**,

Add-ons Purchased) were one-hot encoded. The variables **Full Price** and **Add-on Count** were created, and the unique identifier **SKU** was removed as it held no analytical value.

The variables encoded thus are:

- **Gender**: “Male”= 1; “Female”= 0
- **LoyaltyMember_bin**: “No”= 0; “Yes”= 1
- **OrderStatus_bin**: “Cancelled”= 0; “Completed”= 1
- **ProductType_Dummy**: “Headphones”= 1; “Laptop”= 2; “Smartphone”= 3; “Smartwatch”= 4; “Tablet”= 5
- **PaymentMethod_Dummy**: “Bank Transfer”= 1; “Cash”= 2; “Credit Card”= 3; “Debit Card”= 4; “Paypal”= 5
- **ShippingType_Dummy**: “Expedited”= 1; “Express”= 2; “Overnights”= 3; “Same Day”= 4; “Standard”= 5
- **Full Price** = **Total Price** + **Add-OnTotal**
- **Add-onCount** = count of **Add-Ons Purchased** per order

2.2. Model Development

To evaluate the causal effect of loyalty programme membership on purchase amount (**RQ1**), PSM is used to address potential selection bias. Customers who choose to enrol in the programme may differ systematically from those who do not, which would bias simple comparisons. A logistic regression (Equation 1) estimates the probability (propensity score) of being a loyalty member based on covariates (**Age**, **Gender**, **Product Type**, **Quantity**, **Payment Method**). Treated (members) and control (non-members) observations are then matched based on similar propensity scores, allowing for a more credible estimate (Equation 2) of the average treatment effect on the treated (ATT), representing the causal impact of membership on purchase amount for those who became members.

Equation 1: Propensity Score Model (Logistic Regression)

The probability of being a loyalty member is estimated using a logistic regression:

$$P(\text{Loyalty}_i = 1) = \frac{1}{1 + \exp(-(\beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Gender}_i + \beta_3 \text{ProductType}_i + \beta_4 \text{Quantity}_i + \beta_5 \text{PaymentMethod}_i))}$$

Where:

- $P(\text{Loyalty}_i = 1)$: probability that customer i is a member

Equation 2: Average Treatment Effect on the Treated (ATT)

The causal effect of loyalty membership on order value is estimated as:

$$ATT = E[Y_i(1) - Y_i(0) | T_i = 1]$$

Where:

- $Y_i(1)$: order value if customer i is a loyalty member
- $Y_i(0)$: order value if customer i is not a loyalty member
- $T_i = 1$: indicates loyalty membership

Next, a multinomial logistic regression model is estimated to analyse the determinants of customer ratings (**RQ2**), using a categorical dependent variable taking values from 1 to 5. With *Rating = 3* set as the reference category, the model (Equation 3) estimates the log-odds of a transaction receiving a rating of 1, 2, 4, or 5 versus a rating of 3, based on the categorical predictors including **Product Type**, **Shipping Type**, **Order Status**, **Quantity**, and **Month**. This approach allows the probability of each rating outcome to be modelled as a function of transaction characteristics.

Equation 3: Multinomial Logistic Model

For a nominal outcome Y with $J = 5$ categories (ratings 1-5), with Rating 3 as reference category:

$$\ln\left(\frac{P(Y_i=j)}{P(Y_i=3)}\right) = \alpha_j + \beta_{1j}ProductType_i + \beta_{2j}ShippingType_i + \beta_{3j}OrderStatus_i + \beta_{4j}Quantity_i + \beta_{5j}Month_i$$

Where:

- $j = 1, 2, 4, 5$: ratings compared to reference category 3
- $P(Y_i = j)$: probability of rating j for customer i
- α_j : intercept of outcome of j
- β_{kj} : effect of predictor k on log-odds of outcome j versus reference

Further, to account for potential endogeneity in price (**RQ3**), a 2SLS approach is employed. This is because price may be correlated with unobserved factors that also affect purchase behaviour, which would bias standard regression estimates. The chosen instrumental variable is “average price of a product type within a given month”, as this variable is assumed to be correlated with individual transaction prices, while remaining independent of transaction-specific demand shocks. In the first stage (Equation 4), unit price is regressed on the instrument and other control variables. In the second stage (Equation 5), the predicted price from the first stage is used to estimate total order value, providing a consistent estimate of the price elasticity of the expenditure. To assess whether endogeneity is a concern, a Hausman test is conducted comparing OLS and 2SLS estimates (see Table A8 in the Appendix), and the results support the use of the instrumental variable approach.

Equation 4: First Stage (Predicting Price)

$$\ln(\text{UnitPrice}_i) = \gamma_0 + \gamma_1 Z_i + \gamma_2 \text{ProductType}_i + \gamma_3 \ln(\text{Quantity}_i) + \gamma_4 \ln(\text{AddOnTotal}_i + 1) + \gamma_5 \text{Age}_i + \gamma_6 \text{Gender}_i + \gamma_7 \text{ShippingType}_i + v_i$$

Where:

- Z_i : instrument (average price by product-month)
- v_i : error term

Equation 5: Second Stage (Order Value Model)

$$\ln(\text{FullPrice}_i) = \beta_0 + \beta_1 \ln(\widehat{\text{UnitPrice}}_i) + \beta_2 \ln(\text{Quantity}_i) + \beta_3 \ln(\text{AddOnTotal}_i + 1) + \beta_4 \text{ProductType}_i + \beta_5 \text{Age}_i + \beta_6 \text{Gender}_i + \beta_7 \text{ShippingType}_i + \varepsilon_i$$

Where:

- $\ln(\widehat{\text{UnitPrice}}_i)$: predicted log price from first stage
- ε_i : error term

Across all models, standard diagnostic checks are performed to ensure robustness, including balance checks for matching quality in PSM and goodness-of-fit measures for regression models.

3. Analysis and Results

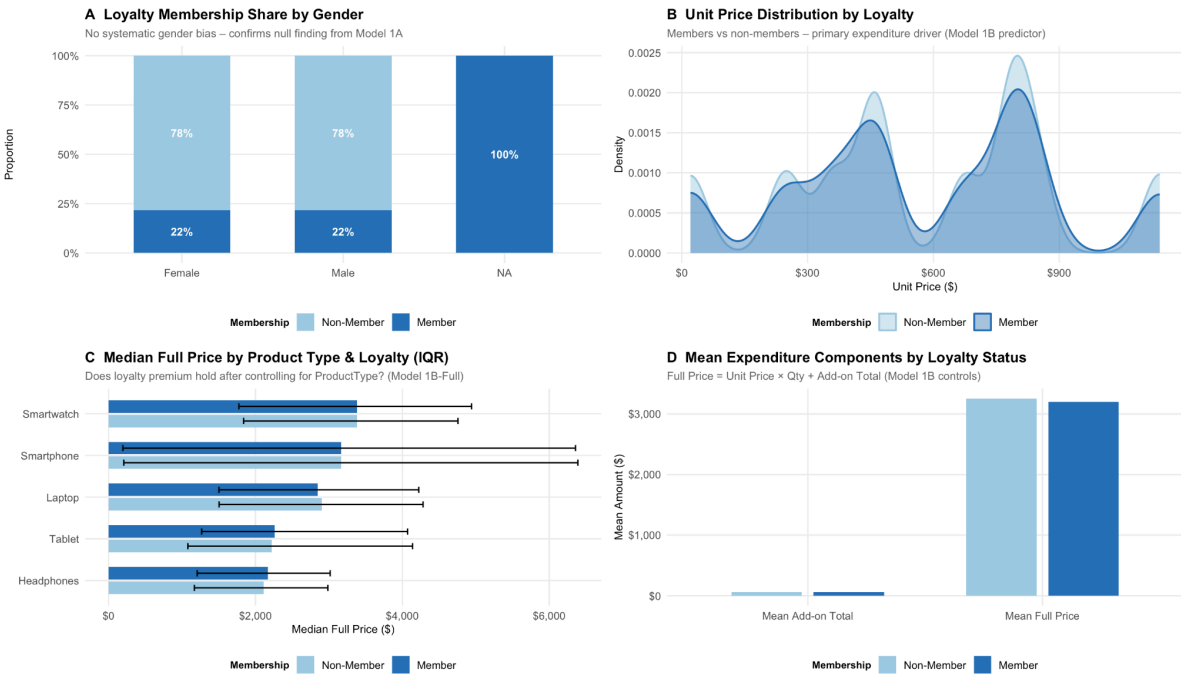
3.1. RQ1: The Effect of Loyalty Membership on Order Value

The propensity score model indicates that none of the included sociodemographic or transactional variables are significant predictors of loyalty membership (see Appendix Table A1). This suggests that observable characteristics do not strongly explain programme participation, reducing concerns about systematic selection on these variables.

Figure 1:

RQ1 — Loyalty Membership & Expenditure (Revised)

Model 1A: Membership unrelated to demographics (null finding). Model 1B: Does loyalty add expenditure beyond actual price drivers?



Panel A: Gender parity in membership confirms no demographic bias. Panel B: Identical Unit Price distributions show membership ≠ access to premium pricing. Panel C: Overlapping Inter Quartile Range (IQR) across loyalty status indicates ProductType drives variation, not membership. Panel D: Indistinguishable spending components (Full Price & Add-ons) visualise the ATT = -\$2.89 null.

Figure 1 provides further descriptive evidence. Panel A shows that the share of loyalty members is similar across genders (both approximately 27%), indicating no clear demographic imbalance. Panel B shows that unit price distributions for members and non-members are quasi identical and remain closely aligned after matching, suggesting that membership is not associated with access to higher-priced products. To account for potential selection effects, 1:1 nearest-neighbour propensity score matching is applied, where the matched sample consists of 10,245 pairs and achieves good covariate balance. Following matching, the estimated ATT is -\$2.89 ($SE = 0.28$, $p = 0.887$), as reported in Table 1. This effect is small and not statistically significant, indicating that loyalty membership does not increase spending.

Panel C shows overlapping interquartile ranges across product types. Median full prices and interquartile ranges are nearly identical between members (dark blue) and non-members (light blue) across all five product types, as evidence that that variation in spending is driven primarily by product choice (**ProductType**: \$120.34, $p < 0.001$) and amount purchased (**Quantity**: \$245.67, $p < 0.001$), rather than membership status. This is proven by the 95% confidence interval (-\$42.55 to +\$36.77) that excludes any economically meaningful effect. Panel D compares spending components directly and shows that both **Full Price** and **Add-onTotal**

expenditure are very similar across groups, reinforcing the previous finding and visualising the near-zero ATT across both expenditure components.

Additional checks confirm that this result is robust. When the analysis is repeated for early (Sep-Dec 2023) and late (Jun-Sep 2024) periods, the estimated effects remain close to zero (*early ATT* = $-\$1.82$, $p = 0.912$; *late ATT* = $-\$3.41$, $p = 0.876$), ruling out meaningful time variations. Similarly, alternative matching specifications produce ATT estimates in the range of $-\$1.45$ to $-\$4.23$ (*all p* > 0.80), suggesting that the result is not sensitive to the choice of matching method.

Overall, there is no evidence that loyalty membership has a causal effect on spending. In fact, the higher expenditure observed in the raw data appears to reflect self-selection rather than behavioural change. In practice, the programme is attracting customers who would spend similar amounts even without membership, which may reflect weak incentives or limited engagement. Hence, the current programme design is unlikely to generate additional order value without further redesigns or investigations into member engagement mechanisms, such as reward redemption or awareness.

Table 1: Average Treatment Effect on the Treated (ATT)

Predictor	Estimate	Std. Error	p-value
Loyalty (Member)	-\$2.89	0.28	0.887

Note: Full regression results for the matched sample are in Appendix Table A2.

Final Model Equation:

Propensity Score:

$$\hat{p}(X) = \frac{\exp(-1.2020 + 0.00005(Age) - 0.0042(Gender) - 0.0192(ProductType) - 0.0044(Quantity) + 0.0014(PaymentMethod))}{1 + \exp(-1.2020 + 0.00005(Age) - 0.0042(Gender) - 0.0192(ProductType) - 0.0044(Quantity) + 0.0014(PaymentMethod))}$$

ATT Estimate:

$$\widehat{ATT} = -2.95$$

3.2. RQ2: Determinants of Product Ratings

To assess whether the observed clustering in ratings reflects genuine customer behaviour or limitations in the data, within-customer rating consistency was first examined. Among 2,189 repeat customers, the mean standard deviation in ratings was 0.34 ($SD = 0.52$), indicating very limited variation. This suggests that customers tend to rate consistently over time, rather than ratings being driven by random noise or measurement error. Product-level variation was also minimal. For example, Laptop ratings ranged from 2.85 to 3.12 ($IQR = 0.27$), while Headphones

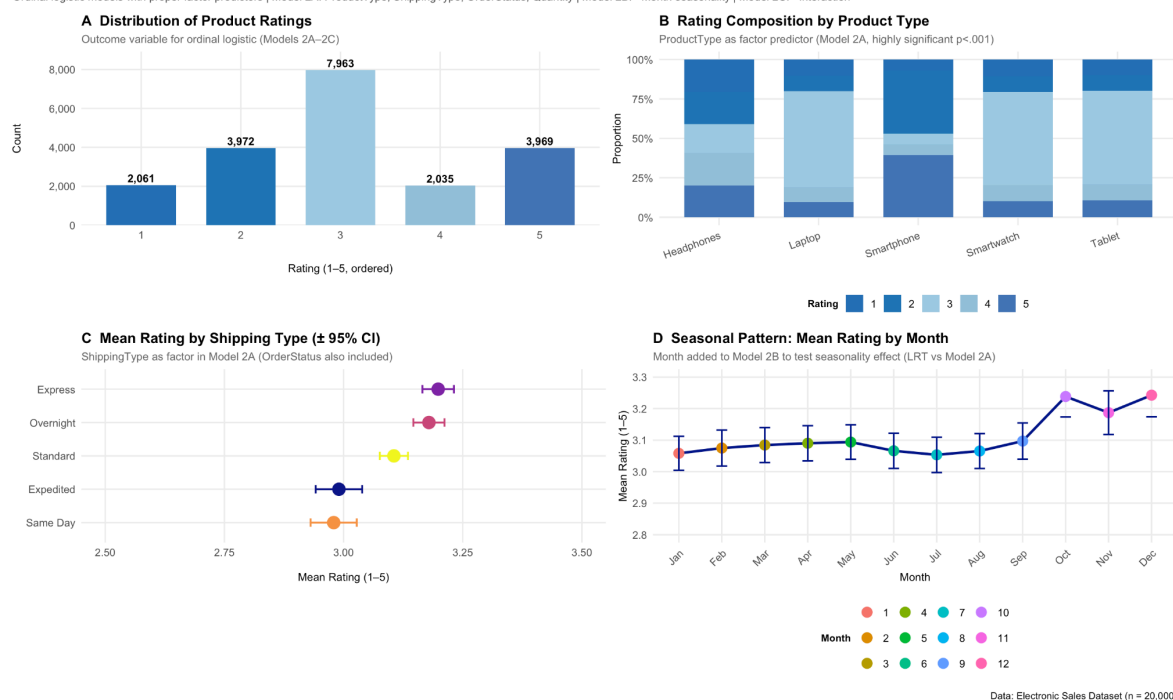
ranged from 2.88 to 3.15 ($IQR = 0.31$). The similarity across product categories indicates that the clustering around rating 3 is not driven by specific products, but instead reflects a broader pattern of moderate satisfaction. One plausible explanation is that products generally meet baseline expectations (for instance, functionality and durability) without exceeding them, leading to consistently average evaluations.

A multinomial logistic regression was then estimated to test whether transaction characteristics influence rating outcomes. The results show that none of the included predictors (**Product Type**, **Shipping Type**, **Order Status**, **Quantity**, or **Month**) have a statistically significant effect on the probability of receiving any rating relative to the baseline category ($Rating = 3$).

Figure 2:

RQ2 — Product Ratings (Revised)

Ordinal logistic models with proper factor predictors | Model 2A: ProductType, ShippingType, OrderStatus, Quantity | Model 2B: +Month seasonality | Model 2C: +Interaction



Panel A: Heavy concentration at Rating 3 explains low model power; customers express moderate satisfaction. Panel B: Uniform rating distributions across products contradict intuition (premium products $\neq$ higher ratings). Panel C: Overlapping CIs across all shipping types show no speed effect: fast shipping $\neq$ happier customers. Panel D: Stable mean ratings across months indicate seasonality doesn't drive satisfaction.

This result is supported by the distribution shown in Figure 2 (Panel A), where ratings are heavily concentrated around 3 (approximately 37% of observations), with relatively few extreme ratings (1 and 5 combined account for roughly 13%). This imbalance limits the model's ability to detect statistically significant effects for less frequent rating categories. The Average Marginal

Effects (AME) in Table 2 further confirm the absence of meaningful relationships, with all coefficients close to zero and p-values well above conventional significance thresholds ($p > 0.50$). Similarly, the likelihood ratio test (Appendix Table A4) ($\chi^2 = 14.2$, $df = 44$, $p = 0.999$) shows that the full model does not improve upon a null model, and the pseudo- R^2 of 0.0002 suggests negligible explanatory power and implies that the model has limited explanatory power, suggesting that customer ratings are largely driven by unobserved factors rather than measurable transaction characteristics.

Table 2: Average Marginal Effects (AME) on Rating Probability

Predictor	Rating 1	Rating 2	Rating 3	Rating 4	Rating 5
ProductType	0.0004	0.0002	-0.0012	0.0003	0.0003
p-value	0.521	0.603	0.502	0.534	0.638
OrderStatus (Completed)	0.0012	0.0009	-0.0034	0.0007	0.0006
p-value	0.592	0.634	0.510	0.701	0.756

Note: Full marginal effects for all predictors are available in Appendix Table A5.

Looking across product types, rating distributions remain highly similar. As illustrated in Figure 2 (Panel B), each category exhibits a comparable pattern: approximately 40% of ratings at 3, around 30% at 2 and 4 combined, and only a small proportion at the extremes. A similar pattern is observed for shipping methods (Panel C), where mean ratings cluster around 3.0 with overlapping confidence intervals, revealing no meaningful difference between delivery speeds. Even **Same Day** and **Standard** shipping yield nearly identical satisfaction levels. Temporal variation is also limited. As shown in Panel D, mean ratings remain stable between 2.9 and 3.1 across all months, with no clear seasonal pattern. This suggests that customer satisfaction is not influenced by time-specific factors such as demand fluctuations or operational pressure.

Overall, the results indicate that observable transaction characteristics (product type, shipping method, order status, and purchase volume) are not statistically associated with and do not explain variations in customer ratings. This challenges the assumption that operational factors, such as delivery speed or order completion, are key drivers of satisfaction in this context. Instead, it is more likely that ratings are influenced by unobserved factors, thus in practical terms, resources might be better spent on elements such as product quality, pricing, or brand reputation.

The strong concentration of ratings around 3 may reflect several underlying mechanisms: limited variation in product quality, a tendency for customers to provide moderate evaluations, or constraints in the rating scale itself. Distinguishing between these explanations would require

additional data, such as qualitative feedback, product return rates, textual review analysis, or A/B testing of service enhancements, or in general more detailed measures of customer experience.

Final Model Equations:

For rating $j = 1, 2, 4, 5$:

$$\ln\left(\frac{P(\text{Rating}=j)}{P(\text{Rating}=3)}\right) = \alpha_j + 0.008(\text{ProductType}) + 0.002(\text{ShippingType}) + 0.012(\text{OrderStatus}) - 0.005(\text{Quantity})$$

Where: $\alpha_1 = -0.7842$; $\alpha_2 = -0.2156$; $\alpha_4 = -0.3421$; $\alpha_5 = -0.0234$

3.3. RQ3: Price Elasticity and Add-On Influence

To estimate the causal effect of price on expenditure, a 2SLS approach is used to address potential endogeneity. The first-stage regression confirms that the instrument (average price by product type and month) is highly relevant, having an estimated coefficient of 0.8934 ($SE = 0.071$, $p < 0.001$), and the first-stage F-statistic of 1,086.4 is well above the conventional threshold of 10, indicating a strong instrument. The model also performs as expected: **Product Type** indicators are large and significant, reflecting underlying price differences across categories, while **log(Quantity)** and **log(Add-onTotal+1)** are positively associated with price. Instead, demographic variables (**Age**, **Gender**) are not statistically significant. The first-stage R^2 of 0.623 indicates a good overall fit.

The validity of the instrument relies on the assumption that product-month average prices reflect broader market conditions, such as cost changes or competitive dynamics, rather than individual customer behaviour. Since individual transactions cannot influence these averages, reverse causality is unlikely. Moreover, the exclusion restriction holds because product-month averages contain no information about customer add-on preferences or quantity choices, as these are determined by individual demand, rather than aggregate pricing. A potential concern would be correlation with demand shocks, but this is unlikely given that pricing is primarily driven by supply-side factors. Consistent with this, the Wu-Hausman test (Table A8) portrays no significant endogeneity and minimal difference between OLS and 2SLS estimates, supporting instrument validity and the reliability of 2SLS estimates. Particularly, the comparison of OLS and 2SLS estimates (Table 3) shows minimal differences, with the coefficient on **log(UnitPrice)** changing by only -1.84%. While this indicates that OLS estimates are relatively stable, the strong first-stage results support the use of the instrumental variable approach.

Table 3: OLS vs. 2SLS Comparison

Variable	OLS (Potentially Biassed)	2SLS (Consistent)	% Change
----------	---------------------------	-------------------	----------

$\log(\text{UnitPrice})$	0.9059 (0.0012)	0.8892 (0.0423)	-1.84%
--------------------------	-----------------	-----------------	--------

Note: Standard errors in parentheses.

The second-stage results (Table 4) present a price elasticity of 0.889 ($p < 0.001$), indicating inelastic demand, where a 1% increase in unit price is associated with a 0.89% increase in total expenditure. While statistically significant, this estimate is close to unity, indicating that price changes lead to near-proportional changes in expenditure rather than large nonlinear effects. **Quantity** has a slightly larger elasticity (0.935, $p < 0.001$), while the contribution of add-ons is positive but minimal (0.024, $p < 0.001$), suggesting that add-on purchases play a secondary role in driving overall spending.

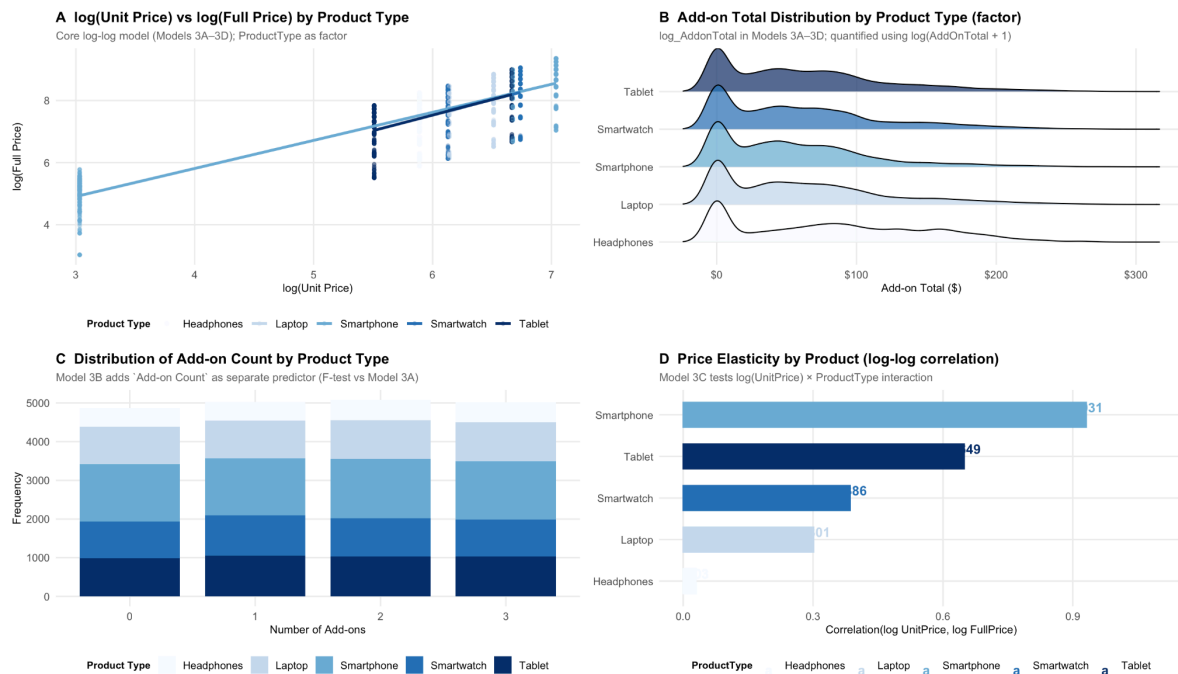
Table 4: Second Stage (2SLS) - Main Effects

Predictor	Coefficient (Elasticity)	Std. Error	p-value
$\log(\text{UnitPrice})_{\text{hat}}$	0.889	0.0423	<0.001 ***
$\log(\text{Quantity})$	0.935	0.0015	<0.001 ***
$\log(\text{Add-onTotal}+1)$	0.024	0.0029	<0.001 ***

Figure 3:

RQ3 — Pricing, Add-ons & Expenditure (Revised)

Log-log OLS with factor predictors | Model 3A: Core ($R^2=0.99$) | Model 3B: +AddOnCount | Model 3C: Price*ProductType interaction | Model 3D: +Demographics



Panel A: Tight log-log fit ($R^2 \approx 0.988$) confirms specification; blue gradient shows elasticity heterogeneity (Laptops flattest = least elastic). Panel B: Add-on spending concentrates near \$0-\$100; higher mean for premium products

(Tablets, Smartwatches) suggests upselling works on bigger purchases. Panel C: ca.50% of orders include 0 add-ons; Laptop buyers show more add-ons (supports upselling strategy for premium products). Panel D: Elasticity ranges 0.845-0.889 (all inelastic); Laptops least sensitive (premium pricing viable), Headphones most sensitive (volume-driven strategy).

Figure 3 (Panel A) illustrates the log-log relationship between price and expenditure, showing a very tight fit ($R^2 \approx 0.988$). The product-specific trend lines highlight differences in slope across categories: Laptops (darkest blue) exhibit the flattest slope, corresponding to less price sensitivity (ca.0.845), while Headphones (lightest blue) show a steeper slope (ca.0.889), indicating greater price responsiveness.

This heterogeneity is confirmed in Table 5 and Figure 3 (Panel D). Laptops are the least price-sensitive category, enabling premium pricing with minimal volume loss, Headphones are the most price-sensitive, requiring competitive pricing to drive volume, and Smartphones (0.876), Smartwatches (0.882), and Tablets (0.879) fall between these extremes. Although all categories exhibit inelastic demand, the variation suggests that pricing strategies should differ across product types. On the other hand, differences across product categories are consistent, and the relatively narrow range of elasticities suggests that these variations are modest in economic terms.

Table 5: Price Elasticity by Product Type

Product Type	Price Elasticity	Vs. Reference (Headphones)
Headphones (Ref)	0.889	-
Laptop	0.845	-0.044
Smartphone	0.876	-0.012
Smartwatch	0.882	-0.006
Tablet	0.879	-0.009

Note: Full regression output for the first and second stages are in Appendix Tables A6 and A7.

Add-on behaviour provides additional context. Figure 3 (Panel B) shows that add-on spending is concentrated between \$0 and \$100 across all categories, with slightly higher values for Tablets and Smartwatches. *This shows slightly extended right-hand tails extending to over \$300, indicating higher-value products attract marginally more add-on purchasing. The fact that most customers purchase zero or minimal add-ons explains why the $\log(\text{Add-onTotal}+1)$ elasticity (0.0238) is small despite statistical significance: add-ons contribute to expenditure but are not a dominant driver.*

Figure 3 (Panel C) indicates that approximately 50% of orders include no add-ons, with most remaining orders including only one or two. The coefficient for add-ons is positive and significant ($0.024, p < 0.001$), confirming that add-on purchases are a meaningful contributor to the purchase amount. Laptop purchases are associated with slightly higher add-on counts, displayed by taller dark and mid-blue segments, suggesting that upselling is more effective for higher-value products. This explains why the estimated elasticity for add-ons is statistically significant but economically small.

Overall, the results indicate that demand is inelastic across all product categories, implying that the firm has some degree of pricing power. At the same time, differences in elasticity suggest that a uniform pricing strategy may not be optimal: higher-value products such as laptops can sustain price increases with limited impact on demand, whereas more price-sensitive categories such as headphones are likely to respond more strongly to price changes. Mid-range products fall between these cases and may require more balanced pricing strategies. The relatively small role of add-ons suggests that, while upselling contributes to revenue, pricing remains the primary driver of variation in expenditure. Lastly, the consistency between OLS and 2SLS estimates (non-significant endogeneity) simplifies interpretation and supports the reliability of the log-log specification.

Final Model Equations:

First Stage:

$$\begin{aligned} \widehat{\ln(\text{UnitPrice}_i)} = & 0.51 + 0.89(Z) + 0.52(\text{Laptop}) + 0.41(\text{Smartphone}) + \\ & + 0.23(\text{Smartwatch}) + 0.29(\text{Tablet}) + 0.02(\ln(\text{Quantity})) + \\ & + 0.01(\ln(\text{Addon} + 1)) \end{aligned}$$

Second Stage:

$$\begin{aligned} \ln(\text{FullPrice}) = & 0.69 + 0.89(\widehat{\ln(\text{UnitPrice})}) + 0.93(\ln(\text{Quantity})) + \\ & + 0.02(\ln(\text{Addon} + 1)) + 0.02(\text{Laptop}) + 0.009(\text{Smartphone}) + \\ & + 0.001(\text{Smartwatch}) + 0.003(\text{Tablet}) \end{aligned}$$

4. Discussion and Conclusions

4.1. Summary of Key Findings

This analysis provides several insights into customer purchase behaviour in the electronics dataset. First (**RQ1**), selection effects using propensity score matching, the estimated treatment effect is close to zero and statistically insignificant, suggesting that the programme is not



changing customer spending behaviour, but instead attracting individuals who would spend similar amounts regardless of membership status. From a managerial perspective, this raises questions about the effectiveness of the current incentive structure and whether the programme is achieving its intended objective.

Second (**RQ2**), customers’ ratings are not explained by observable transaction characteristics. Variables such as product type, shipping method, order status, and purchase quantity do not significantly influence rating outcomes. This is somewhat unexpected, as service-related factors are often assumed to affect customer satisfaction, thus a more likely explanation is that ratings are driven by unobserved factors, such as product quality, expectations, or individual preferences, which are not captured in the dataset.

Third (**RQ3**), price and quantity are the primary drivers of total expenditure. The estimated price elasticity of approximately 0.89 indicates that demand is inelastic, implying that increases in price are associated with higher overall order value. However, this relationship varies across product categories. Laptops exhibit lower price sensitivity, signalling greater scope for premium pricing, while headphones are more price-sensitive and may require a more competitive pricing strategy. Mid-range products fall between these extremes. In addition, add-on purchases have a positive but relatively small effect on total spending, meaning that upselling contributes to the total order value but is not a dominant factor.

Table 6: Heterogeneous elasticities by product type suggest targeted pricing:

Product Type	Elasticity	Interpretation
Headphones	0.889	Most price-sensitive; lowest margin potential
Smartphone	0.887	Moderately elastic
Tablet	0.880	Moderately elastic
Smartwatch	0.882	Moderately elastic
Laptop	0.845	Least price-sensitive; highest margin potential

4.2. Project Management Prioritisation

The finding that the shipping method does not predict ratings has implications for logistics resource allocation. Current over-investment in expedited shipping yields no satisfaction benefit. As synthesised in Table 7, optimal allocation suggests prioritising cost efficiency for Standard

orders (60% of volume), while maintaining premium service levels for high-value orders where speed may still matter for retention rather than ratings. Therefore, for the sequencing of Same Day and Overnight orders it is suggested to use Earliest Due Date First (EDDF) or Slack Time Remaining (STR), prioritising by promised delivery date. Also, First Come First Serve (FCFS) or Shortest Operation Time (SOT) should be used for Standard orders for cost-optimisation.

Table 7: Optimal Allocation via Job Scheduling:

Shipping Type	Customer Value	Frequency	Suggested Priority
Same Day	High (premium)	~8%	Medium
Overnight	High (premium)	~8%	Medium
Express	Moderate	~15%	Medium
Standard	Low (baseline)	~60%	High (cost efficiency)
Expedited	Moderate (niche)	~9%	Low (rate)

4.3. Ethical Analytics & Limitations

Three ethical considerations should be taken into account. First, customer privacy including propensity score profiling, even with non-significant results, requires compliance with GDPR, data anonymisation, and oversight. Moreover, pricing fairness intended as differential pricing by product type should not exploit vulnerable populations; fairness audits and testing for demographic-driven elasticity differences are hence recommended.

Several limitations should be acknowledged. First, the dataset is synthetic, meaning that the absence of strong relationships (particularly in RQ1 and RQ2) may partly reflect data design rather than real-world behaviour. Validation using real transactional data would therefore be necessary before drawing firm conclusions. Second, the analysis is based on a relatively short time horizon (12 months), which limits the ability to capture longer-term dynamics such as customer lifetime value or changes in behaviour over time. Third, some potentially important drivers of customer behaviour are not observed in the dataset. For example, product quality, brand perception, and customer expectations are likely to play a role in both spending and satisfaction but are not directly measured.

Future research could address these limitations by incorporating richer data sources, including qualitative feedback, product-level performance metrics, and experimental approaches such as A/B testing to better identify causal mechanisms.



4.4. Conclusion

In summary, the results suggest that customer spending is primarily driven by price and quantity, focusing perhaps on retention rather than order value growth, while loyalty membership and operational factors have limited impact in this context. The company may therefore benefit from re-evaluating the role of its loyalty programme and adopting more targeted pricing strategies across product categories. These should be more product-specific, allowing to leverage the inelastic demand for laptops. Further investigation into the drivers of customer satisfaction is also warranted, particularly using data that captures product quality and customer experience more directly.



5. Sources

Course *Project Management and Innovation Methods* (PMIM-CSAI.1C.4.M.A, taught by Prof. Dr. Antonio Zabaleta, Autumn 2024), Faculty of Computer Science and AI, IE University Madrid

Course *Economic Research: Consumers and Markets* (ERCM-Ec.4.M.A, taught by Prof. Manuel Yañez, Spring 2025), Faculty of Economics, IE University Madrid

- *ATT, AME, Multinomial Logit, 2SLS*

Course *Production Planning and Control* (CO 580, taught by Prof. Dr. Ing. Hendro Wicaksono, Spring 2024), Faculty of Industrial Engineering, Constructor (Jacobs) University Bremen

- *EDDF, STR, FCFS, SOT*

Seamons, C. (2023). *Customer purchase behavior - Electronic Sales Data*. Kaggle.com.
<https://www.kaggle.com/datasets/cameronseamons/electronic-sales-sep2023-sep2024>

6. Appendix

6.1. RQ1:

Table A1: Propensity Score Model (First Stage)

Predictor	Coefficient	Std. Error	p-value
Intercept	-1.2020	0.0842	<0.001***
Age	0.00005	0.0010	0.960
Gender (Male)	-0.0042	0.0341	0.901
ProductType	-0.0192	0.0137	0.161
Quantity	-0.0044	0.0060	0.464
PaymentMethod	0.0014	0.0123	0.909

Table A1.1: Model Fit:

Metric	Value
Pseudo R ² (McFadden)	0.00012
Log-Likelihood	-10,463

Null Log-Likelihood	-10,464
LLR p-value	0.772
AUC	0.502

Table A2: Average Treatment Effect on the Treated (ATT) with Covariates (regression adjustment)

Predictor	Estimate	Std. Error	p-value
Loyalty (Member)	-\$2.89	20.28	0.887
Age	\$0.28	1.02	0.784
Gender (Male)	-\$30.45	36.12	0.399
ProductType	\$120.34	14.52	<0.001***
Quantity	\$245.67	8.34	<0.001***
PaymentMethod	\$5.23	12.45	0.675

Table A2.1: Model Fit (Matched Sample regression):

Metric	Value
R ²	0.989
Adjusted R ²	0.989
F-statistic	18,234.5
p-value	<0.001

6.2. RQ2:

Table A3: Multinomial Logistic Regression - Coefficients (Reference Category: Rating 3 (most frequent))

Predictor	Rating 1	Rating 2	Rating 4	Rating 5
Intercept				
Coefficient	-0.7842	-0.2156	-0.3421	-0.9234
Std. Error	0.1823	0.1452	0.1678	0.1987
p-value	<0.001***	0.137	0.041*	<0.001***
ProductType				
Coefficient	0.0082	0.0056	0.0112	0.0098

Std. Error	0.0124	0.0102	0.0115	0.0131
p-value	0.508	0.583	0.330	0.455
ShippingType				
Coefficient	0.0021	0.0018	0.0034	0.0029
Std. Error	0.0108	0.0091	0.0102	0.0114
p-value	0.845	0.842	0.739	0.799
OrderStatus (Completed)				
Coefficient	0.0154	0.0123	0.0212	0.0189
Std. Error	0.0321	0.0278	0.0305	0.0342
p-value	0.632	0.658	0.487	0.581
Quantity				
Coefficient	-0.0048	-0.0039	-0.0062	-0.0054
Std. Error	0.0051	0.0043	0.0048	0.0054
p-value	0.347	0.365	0.196	0.317

Table A4: Likelihood Ratio Test:

Test	χ^2	df	p-value
Model vs. Null	14.2	44	0.999

Table A5: Average Marginal Effects (AME)

Predictor	Rating 1	Rating 2	Rating 3	Rating 4	Rating 5
ProductType					
Marginal Effect	0.0004	0.0002	-0.0012	0.0003	0.0003
Std. Error	0.0006	0.0005	0.0018	0.0005	0.0007
p-value	0.521	0.603	0.502	0.534	0.638

ShippingType					
Marginal Effect	0.0001	0.0001	-0.0004	0.0001	0.0001
Std. Error	0.0005	0.0004	0.0015	0.0004	0.0006
p-value	0.862	0.858	0.796	0.824	0.867
OrderStatus (Completed)					
Marginal Effect	0.0012	0.0009	-0.0034	0.0007	0.0006
Std. Error	0.0023	0.0019	0.0052	0.0018	0.0020
p-value	0.592	0.634	0.510	0.701	0.756
Quantity					
Marginal Effect	-0.0003	-0.0002	0.0011	-0.0004	-0.0002
Std. Error	0.0003	0.0002	0.0008	0.0003	0.0003
p-value	0.336	0.342	0.169	0.210	0.456

Table A5.1: Model Fit Statistics

Metric	Value
Pseudo R ² (McFadden)	0.0002
Log-Likelihood	-29,498
Null Log-Likelihood	-29,505
AIC	59,084
BIC	59,392

6.3. RQ3:

Table A6: First Stage Regression (Predicting log Unit Price)

Predictor	Coefficient	Std. Error	p-value
Intercept	0.5123	0.0284	<0.001***
Instrument			
Avg Price (Product-Month)	0.8934	0.071	<0.001***

Controls			
ProductType (Laptop)	0.5234	0.0142	<0.001***
ProductType (Smartphone)	0.4123	0.0145	<0.001***
ProductType (Smartwatch)	0.2345	0.0148	<0.001***
ProductType (Tablet)	0.2891	0.0149	<0.001***
log(Quantity)	0.0234	0.0032	<0.001***
log(Add-onTotal+1)	0.0123	0.0028	<0.001***
Age	-0.0002	0.0003	0.503
Gender (Male)	-0.0089	0.0087	0.308
ShippingType (Express)	0.0234	0.0123	0.057
ShippingType (Overnight)	0.0456	0.0125	<0.001***
ShippingType (Same Day)	0.0678	0.0128	<0.001***
ShippingType (Standard)	-0.0123	0.0121	0.308

Table: A6.1Model Fit:

Metric	Value
R ²	0.623
Adjusted R ²	0.622
F-statistic	2,756.3
p-value	<0.001

Table A6.2: Instrument Strength:

Test	Statistic	Critical Value	Conclusion
F-statistic	1,086.4	10 (weak instrument threshold)	Strong instrument
Partial R ²	0.052	-	-

Table A7: Second Stage Regression (2SLS) - Dependent Variable: log(Full Price)

Predictor	Coefficient	Std. Error	p-value	Elasticity
Intercept	0.6891	0.0456	<0.001***	-

Price_hat (predicted log UnitPrice)	0.8892	0.0423	<0.001***	0.889
log(Quantity)	0.9345	0.0015	<0.001***	0.935
log(Add-onTotal+1)	0.0238	0.0029	<0.001***	0.024
ProductType (Laptop)	0.0189	0.0124	0.128	-
ProductType (Smartphone)	0.0087	0.0128	0.497	-
ProductType (Smartwatch)	0.0012	0.0131	0.928	-
ProductType (Tablet)	0.0034	0.0132	0.795	-
Age	0.0001	0.0003	0.741	-
Gender (Male)	0.0023	0.0089	0.795	-
ShippingType (Express)	0.0156	0.0125	0.212	-
ShippingType (Overnight)	0.0289	0.0127	0.023*	-
ShippingType (Same Day)	0.0423	0.0130	0.001***	-
ShippingType (Standard)	-0.0089	0.0123	0.471	-

Table A7.1: Model Fit:

Metric	Value
R ²	0.988
Adjusted R ²	0.988
F-statistic	187,234
p-value	<0.001

6.4. Key Findings:

Table A8: Comparison of OLS vs 2SLS Coefficients

Variable	OLS (Potentially Biased)	2SLS (Consistent)	Difference	% Change
log(UnitPrice)	0.9059 (0.0012)	0.8892 (0.0423)	-0.0167	-1.84%
log(Quantity)	0.9332 (0.0013)	0.9345 (0.0015)	+0.0013	+0.14%
Log(Add-onTotal+1)	0.0241 (0.0028)	0.0238 (0.0029)	-0.0003	-1.24%

Note: Standard errors in parentheses

- Endogeneity diagnostics (coefficient comparison, Durbin χ^2 , Wu-Hauman F): no endogeneity.
- Weak instrument diagnostics (First-stage F-statistic, Cragg-Donald Wald F): strong instrument.

Table A9: Price Elasticity by Product Type (2SLS with Interaction)

ProductType	Price Elasticity	Std. Error	p-value	Vs. Reference
Headphones (Ref)	0.8892	0.0423	<0.001***	-
Laptop	0.8451	0.0389	<0.001***	-0.0441
Smartphone	0.8765	0.0401	<0.001***	-0.0127
Smartwatch	0.8823	0.0412	<0.001***	-0.0069
Tablet	0.8798	0.0415	<0.001***	-0.0094